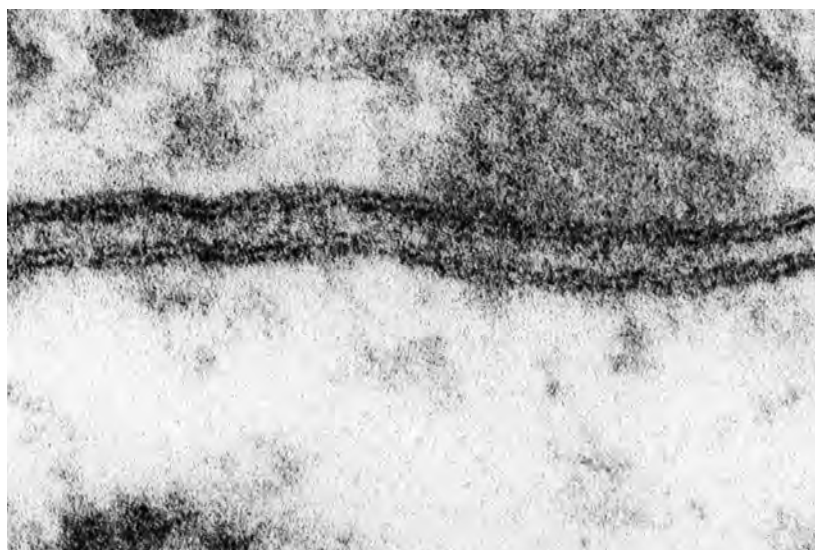


- 5 The structure and properties of the cell membrane are important in controlling which molecules can enter and leave a cell.

- (a) The photograph shows part of a cell membrane, as seen using an electron microscope.



} cell membrane

(Source: © Dennis Kunkel Microscopy / Science Photo Library)

The width of this cell membrane is 5.00 nm.

A phosphate head of a phospholipid is between 0.8 and 0.9 nm in diameter and the fatty acid tails are between 1.25 and 1.75 nm long.

Explain how this electron micrograph provides evidence for the structure of the cell membrane.

Use the information given to support your answer.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

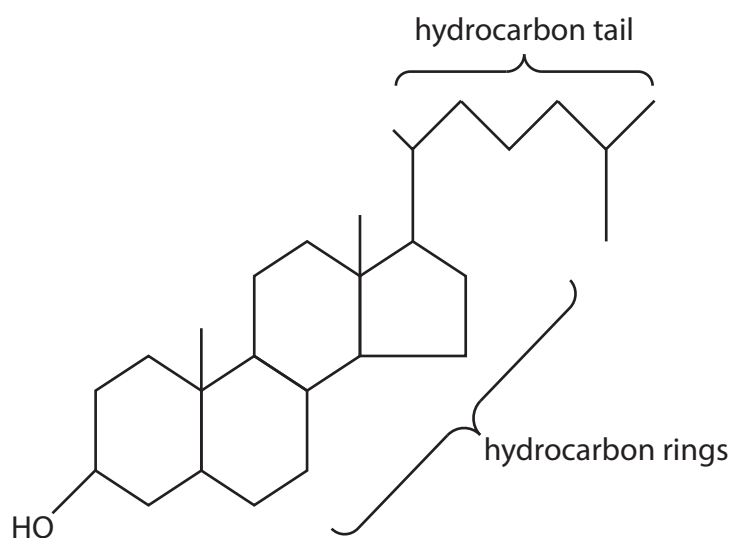
.....



(b) Cell membranes contain cholesterol.

The molecular formula of cholesterol is $C_{27}H_{46}O$.

The diagram shows a cholesterol molecule.



Explain the location of cholesterol in cell membranes.

Use the information in the diagram to support your answer.

(3)

.....

.....

.....

.....

.....

.....

.....

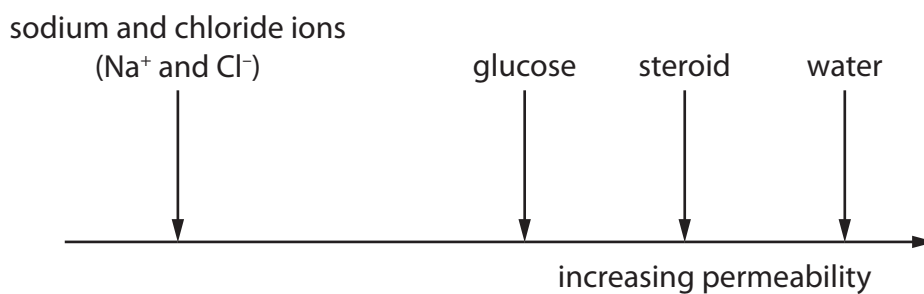
.....

.....

.....



(c) The diagram shows the permeability of cell membranes to some chemicals.



The sodium and chloride ions, glucose and water are polar chemicals.

A steroid is a non-polar chemical.

(i) Describe the dipolar nature of water.

(2)



- (ii) Explain the permeability of cell membranes to each chemical shown in the diagram.

(4)

(Total for Question 5 = 12 marks)

